>_ CYBER TECH ACADEMY

Digital Forensics & Incident Response

Preserve evidence, build timelines, respond to incidents, and verify recovery using disciplined procedures.

LEVEL: ADVANCED

Creator: Dhanush | **Edition:** 1.0 | **Cost:** Free

How to use this guide

Read the concepts first, run examples only in a system you own or are authorised to use, complete the lab, then answer the review questions without looking back. Commands and outputs can differ by Linux distribution, software version, and local configuration.

Responsible learning: only test systems for which you have explicit permission. These notes are educational and do not grant permission to access, scan, disrupt, or modify third-party systems.

Learning objectives

- Explain evidence integrity, chain of custody, and forensic copies.
- Use cryptographic hashes to compare files.
- Build a timeline from multiple event sources.
- Apply a structured incident-response lifecycle.
- Plan containment and recovery while preserving useful evidence.

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1. Incident response foundations

An incident is an event that threatens confidentiality, integrity, or availability and requires a coordinated response. Not every alert is an incident. Triage evaluates evidence, impact, scope, and confidence before choosing actions.

A practical lifecycle includes preparation; detection and analysis; containment; eradication; recovery; and post-incident improvement. These stages can overlap. Organisations should define roles, decision authority, escalation paths, communication plans, and evidence-handling procedures before an incident occurs.

Response decisions balance limiting harm with preserving evidence and maintaining essential services. In a severe incident, safety, legal obligations, business continuity, and specialist advice may take priority. Document decisions and who authorised them.

2. Evidence preservation and chain of custody

Digital evidence can change simply by opening a file, booting a system, or running a tool. A forensic workflow minimises unnecessary changes, documents the acquisition method, records time and operator, and preserves an original or validated forensic copy.

Chain of custody records who collected, handled, transferred, stored, and examined evidence, when each action occurred, and why. Storage should be access-controlled and protected against modification. Use approved procedures and qualified support when legal or regulatory requirements apply.

Prioritise volatile evidence only when the incident plan and expertise support doing so. Memory and live-state collection can be valuable but may change the system. Document tools, versions, commands, timestamps, and limitations.

3. Hashes and file integrity

A cryptographic hash produces a fixed-length digest from input data. SHA-256 is commonly used to compare whether two byte sequences match. If two files have the same secure hash, that is strong evidence they are identical for practical integrity checks; if hashes differ, the contents differ.

```
sha256sum evidence-copy.bin
sha256sum reference-copy.bin
```

Hashing does not prove that a file is benign, authentic, or correctly acquired. A malicious file can have a valid hash. Record where the reference hash came from and protect the record itself.

Do not write to the evidence file while analysing it. Work on a verified copy, keep the original read-only where feasible, and document every transformation.

4. Timeline analysis

A timeline combines events from sources such as authentication logs, endpoint alerts, file metadata, web logs, and cloud audit records. Normalise timestamps carefully and preserve original timezone information. Correlation can reveal a sequence that is not obvious from one log alone.

Consider clock drift, delayed ingestion, duplicated records, retention limits, and gaps. Separate confirmed facts from inferred sequence. For example, a file modification timestamp may be altered or reflect a copy operation rather than the original creation time.

Use a table with timestamp, source, event, confidence, and interpretation. A strong timeline helps answer what happened, when, which assets were affected, and what evidence supports each conclusion.

5. Containment, eradication, and recovery

Containment aims to prevent further harm. Depending on the case, it may include isolating a device, revoking tokens, disabling compromised credentials, blocking confirmed indicators, or restricting network paths. The action must be proportionate and authorised.

Eradication removes the cause, such as malicious persistence, exploited weaknesses, or unauthorised access. Recovery restores trusted systems, rotates affected credentials, validates configurations, and monitors for recurrence. Restoring from backups without removing the root cause may lead to reinfection.

Communication is part of response. Keep concise status updates, identify decision owners, protect sensitive information, and coordinate with legal, privacy, HR, communications, and service owners as appropriate.

6. Post-incident review

A blameless review examines how systems and processes behaved, not just who made a mistake. Document the timeline, impact, root causes, successful controls, missed detections, response friction, and recovery performance.

Turn findings into owned actions with deadlines and measurable outcomes: improve alert coverage, reduce excessive privileges, test backups, patch vulnerable services, or rehearse a playbook. Track completion and retest controls.

Preserve reports and evidence according to retention policies. Share lessons in a way that improves resilience without exposing secrets or unnecessary personal data.

Guided practical lab

Scenario: a fictional company reports suspicious logins and an unexpected archive download from a workstation.

- Record initial facts without assuming who is responsible.
- Identify which logs and endpoint records should be preserved.
- Document evidence owner, collection time, source, and hash where appropriate.
- Recommend containment steps and explain possible business impact.
- Build a timeline that distinguishes confirmed events from hypotheses.
- List eradication, recovery, and post-incident validation actions.

This is a tabletop exercise. Do not acquire or inspect real devices or accounts unless you have the organisation's explicit authority and approved procedures.

Knowledge check

Try to answer these before checking the answer key. Write your reasoning, not just a single word.

1. What is chain of custody?

Answer: _____

2. Does a matching SHA-256 hash prove a file is safe?

Answer: _____

3. Why work from a verified copy?

Answer: _____

4. What is the purpose of containment?

Answer: _____

5. Why must timelines distinguish fact from inference?

Answer: _____

Answer key and reasoning

1. A documented record of who handled evidence, when, how, and for what purpose.

2. No. It supports integrity comparison, not a judgement that the file is benign or trustworthy.

3. It helps preserve the original evidence and makes analysis more repeatable.

4. To limit further harm or spread while the response team investigates and plans remediation.

5. It prevents assumptions from being reported as established evidence and improves decision quality.

Glossary

Term	Meaning
Forensic image	A structured copy of digital media acquired for examination.
Hash	Fixed-length digest used to compare data.
Chain of custody	Documented history of evidence handling.
Containment	Limiting the spread or impact of an incident.
Eradication	Removing the cause or persistence of compromise.
Timeline	Chronological reconstruction from multiple evidence sources.

Further study

Continue with the next Cyber Tech Academy module, practise in a disposable virtual machine or an explicitly authorised lab, and consult official manuals for the tools you use. Verify commands against your own system before running them.

Cyber Tech Academy • Free learning • Learn the theory. Practise safely. Build real skills.